

Radar.lua

1. Setup guide
 - 1.1. Set up a monitor. Suggest 3x3. The radar supports a responsive design so you can have it as large as you want.
 - 1.2. Place the computer containing the programme.
 - 1.3. Place a some peripherals' radar next to the computer
 - 1.4. Run the programme
 - 1.5. The image should appear on screen.
 - 1.6. If the bearing seems disoriented, that's probably because you are assembling the ship in a different direction. Try to assemble the ship in a different way.
2. Radar image reading
 - 2.1. There will be grid lines on the screen. You can see the scale at the bottom left corner. The segment length will be displayed. A segment is the line between 2 intersection points. You can use that to estimate the distance between targets
 - 2.2. At the bottom left corner, there is a scale: -/+ button. Press that to zoom in and out of the map. Pressing the + increase the actual segment length, it increases the total radar range (sort of), making more targets appear inside the circle. So pressing + zoom out. Vice versa for the -
 - 2.3. There will be target that is capped on the circle. The letter represents them will be displayed in lightBlue. This means that the target is out of the current map view distance. Eg, you set the scale to 50m/segment, then target with 100+ distance will in lightBlue.
 - 2.4. There is a compass to show the brief direction of targets and the true bearing of the ship's heading.
3. Threat level indication
 - 3.1. There are 4 types of object displayed.
 - 3.2. Blue: friendly, identified with the predefined friendly id list
 - 3.3. Yellow: Enemy light weight and slow target, eg speedboat
 - 3.4. Orange: Enemy large vessel, eg destroyer
 - 3.5. Red: Enemy fast flying object, eg missile, fighter jets
 - 3.6. The threat level is only determined by mass, velocity, and friendliness. So, it may not represent the actual threat level.
4. Target detail display
 - 4.1. The target's details can be displayed by right clicking the target on the screen.
 - 4.2. There will be numeric details of the target's information.
5. Waypoint
 - 5.1. This can be used with the helmGen2 to plot a cruise map.

- 5.2. Right click the waypoint button.
- 5.3. Open the computer terminal and input as instructed. Press enter to quit
- 5.4. Note that the colors have to be inputed like the colors in computer craft. Example
input “-20,60,lightBlue”, “0,0,white”
- 5.5. You can also right click the waypoint to show more details.

HelmGen2.lua

1. Setup
 - 1.1. Connect a monitor (3*2) a ender modem(on top) and a lectern with tweaked controller.
 - 1.2. Setup the engineControl.lua. The left engine rotational speed controller should be on the left on the computer and right on the right. Connect the computer to the stressmometer, fuel silo and energy accumulator (for APU).
 - 1.3. Run engineControl.lua and helmGen2.lua on the respective computer
2. Display
 - 2.1. There will be a lot of ship operation data displayed.
3. Throttle and turing
 - 3.1. The throttle and turning can be increased and decreased by touching the screen or by using the tweaked controller. W and S(yAxis) is binded for throttle. A and D(xAxis) is binded for turning. For throttle, positive RPM = going forward, vice versa. And for turning, negative is left and positive is right.
4. Waypoint and Auto-pilot
 - 4.1. Click the add waypoint button to add waypoint.
 - 4.2. Open the computer terminal and follow the instructions. Example input “ - 10,20 ”.
 - 4.3. Press enter and you will find the waypoint displayed on the screen.
 - 4.4. When you add another way point, the way point will be marked as waypoint 2...and so on.
 - 4.5. You can press the delete button to delete the waypoint.
 - 4.6. The auto-pilot system will automatically adjust the ship's yaw to head to the waypoint.
 - 4.7. When reaching within 200 blocks of the waypoint, the waypoint will be automatically deleted and the next waypoint will be tracked.